Andhra University College of Engineering

Computer Science and Engineering | Course Code: CS301

Operating Systems - Comprehensive Syllabus Lecture Notes

Syllabus Focus: CS301 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Operating Systems. Includes detailed unit-by-

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Process Management & CPU Scheduling

" Syllabus Scope: Process states, Process Control Block PCB, Context switching overhead, Scheduling algor

Unit 2: Inter-Process Communication & Synchronization

" Syllabus Scope: Critical section problem, Peterson solution, Hardware test-and-set, Counting and binary

CS301: Operating Systems - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Deadlocks & Mitigation Strategies

" Deadlock conditions, Resource allocation graphs, Bankers algorithm for safety and avoidance, Detection algor

Unit 4: Memory Management & Demand Paging

" Logical vs physical address space, Contiguous allocation, Paging, Inverted page tables, Segmentation, Demand

Unit 5: File Systems & Storage Subsystems

" File concepts, Directory structures, Inode allocation methods contiguous, linked, indexed, Free-space manage

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Derive the average turnaround time and waiting time for SJF and Round Robin CPU scheduling on a specif
2. Prove the correctness of Peterson solution for two-process mutual exclusion and verify all three synch
3. Execute the Bankers Algorithm safety routine on a multi-resource allocation matrix and determine wheth
4. Explain demand paging, page fault service routine, and calculate Effective Access Time EAT under varyi

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Abraham Silberschatz, Peter B. Galvin - Operating System Concepts Wiley 10th Ed
* Andrew S. Tanenbaum - Modern Operating Systems Pearson 4th Ed
* William Stallings - Operating Systems: Internals and Design Principles Pearson

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